

Wizard of Oz Prototyping for Interactive Spatial Augmented Reality in HCI Education

Experiences with Rapid Prototyping for Interactive Spatial Augmented Reality

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ABSTRACT

In this paper we present our findings that emerged from Wizard of Oz (WOz) prototyping workshops of an undergrad course on UX design between 2017 – 2022. The purpose of these workshops is both to familiarize the students with WOz as a rapid low-complexity prototyping method and to facilitate them in designing for interactive systems beyond typical graphical user interfaces. In these workshops, students develop and test Wizard of Oz prototypes for interactive spatial augmented reality (SAR). Over the past six years, in total 93 prototypes have been created on the course. We analyzed the prototypes that emerged from the workshops based on input, purpose, play characteristics and projection surfaces. Based on the analysis and our experiences during the workshops, we conclude that the workshops stimulate creativity, enable students to rapidly create interactive SAR interfaces without being limited by their technical skills and time limitations and provide them with a tool that they apply beyond the scope of the prototyping course. We discuss advantages and limitations of the use of WOz prototyping for interactive Spatial Augmented Reality.

CCS CONCEPTS

• **Social and professional topics** → Professional topics; Computing education; Computing education programs; Information systems education; • **Human-centered computing** → Interaction design; Interaction design process and methods; Interface design prototyping.

KEYWORDS

Prototyping, User Experience Design, Interface Design, Design Methods, Human Computer Interaction, Wizard of Oz, Projection

Mapping, Spatial Augmented Reality, Education, Playful Interaction, Emerging Technologies

ACM Reference Format:

Danića Mast, Alex Roidl, and Antti Jylha. 2023. Wizard of Oz Prototyping for Interactive Spatial Augmented Reality in HCI Education: Experiences with Rapid Prototyping for Interactive Spatial Augmented Reality. In *Extended Abstracts of the 2023 CHI Conference on Human Factors in Computing Systems (CHI EA '23)*, April 23–28, 2023, Hamburg, Germany. ACM, New York, NY, USA, 10 pages. <https://doi.org/10.1145/3544549.3573861>

1 INTRODUCTION

In the past 15 years, user experience design (UXD) has evolved from focusing primarily on graphical user interfaces (GUIs) to a broader design space. The emergence of mainstream technologies for advanced motion tracking, speech recognition a.o., have brought new kinds of products into the market and new challenges for designing digital experiences. While advanced prototyping tools and techniques for GUIs are in the toolbox of every UX designer, newer and more complex domains such as interactive spatial augmented reality (SAR) and voice interfaces require different approaches in prototyping.

Wizard of Oz (WOz) prototyping is a method that allows the evaluation of interactive systems before fully implementing their autonomous elements and is very useful for designing and evaluating technologically complex interfaces. The interactivity in the system is controlled by a hidden human operator (the Wizard).

In the UX design undergraduate program at The Hague University of Applied Sciences, we aim at educating UX designers who have the skills and tools to design digital products that go beyond screen-based interactions. Therefore, during the course module Building Engaging Prototypes, we teach the WOz prototyping approach to our first-year students. The assignment, facilitated through two sequential workshops, is to design and test a WOz prototype for an interactive SAR installation. During the years 2017 – 2022, we have conducted these workshops with six cohorts of students, which has resulted in 93 WOz prototypes. In this paper, we discuss the method of teaching WOz as a rapid, low-barrier prototyping tool; the characteristics of the prototypes in terms of

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CHI EA '23, April 23–28, 2023, Hamburg, Germany

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ACM ISBN 978-1-4503-9422-2/23/04.

<https://doi.org/10.1145/3544549.3573861>

input, purpose, play characteristics and projection surfaces; and the effectiveness of the method to stimulate the creativity of aspiring designers.

2 BACKGROUND

2.1 History of WOz Prototyping

Wizard of Oz prototyping is a powerful and useful experience prototyping method for testing interface design ideas at an early stage in the design process. WOz prototypes enable a user to interact with a system that (unknowingly to the user) is controlled by a hidden human operator (the wizard). The wizard controls the system instead of a fully functioning technological backend. This gives the user the illusion that the system works autonomously.

WOz prototypes allow designers, developers and researchers to quickly develop and test interface ideas in the early stages of designing systems such as ambient interfaces, interactive installations, and speech-based interactions, without having to invest a lot of time and money building a fully working interactive prototype. Furthermore, it allows testing of technology that is not yet available or working properly but might be in the (near) future.

The technique in the context of HCI emerged around 1975 in Chapanis' Communications Lab at Johns Hopkins University [3] and was initially referred to as 'experimenter in the loop technique'. The term 'Wizard of Oz' was first used around 1980 [10], initially referred to as the OZ paradigm [8, 9]. However, the use of a human operator to simulate a technical backend goes back as far as the late 18th century. An infamous example is the Mechanical Turk (aka. Maelzel's Chess Player) [13], that was exhibited as an autonomous chess-playing automaton, although it was actually controlled by a human chess player.

Nowadays, Wizard of Oz prototyping is used as a prototyping method in psychology, linguistics and within HCI for spoken or typed natural language systems [1], a.o. Wizard-of-Oz-like methods are also used as part of companies' business strategies. For example, the Aardvark company developed a social search system, initially operated by humans to replicate an autonomous backend as much as possible, thereby gathering data that enabled them to further develop their backend technology [15].

2.2 WOz in HCI Education

Within HCI Education, WOz prototyping is part of many HCI-related curricula. However, a review of course syllabi shows that WOz prototyping is taught as a possible method, but rarely required as method for course assignments.

Besides the many advantages of WOz prototyping, this method also has shortcomings. It has a high cognitive load for the Wizard, there are few tools supporting WOz prototyping specifically [6] and it requires development of both the prototype interface and Wizard controls. Attempts to better facilitate WOz prototyping are WOz pro [6] and Ex-A-Sketch [4]; both projects that have emerged from teaching WOz prototyping in an educational context.

2.3 Wizard of Oz & Interactive Spatial Augmented Reality

Spatial Augmented Reality (SAR), aka. Projection Mapping (Mast et al. 2017), entails projecting digital content onto real-worlds three-dimensional surfaces and objects [11, 12]. WOz prototyping is a

helpful prototyping method for developing interactive SAR. These systems often have a complex technical backend, because they use, for example, computer vision technology or tangible electronics as input methods. This makes it worthwhile to apply WOz prototyping early in the research and development process to make the process more efficient and affordable by avoiding the need to first develop a completely working system [2]. Examples of WOz being used for SAR prototyping include the development of an augmented climbing wall [7], touchless gestural interaction [5], and collaborative learning on tabletops [14].

3 METHOD

3.1 Structure of the Workshops

Wizard of Oz prototyping is one of the methods first-year User Experience Design B.Sc. students learn during the eight-week course

You will make a Wizard of Oz prototype as explained [...] Make an interactive application without writing (much) code and simulate machine behaviour with human operators.

The prototype should:

- *work with a video projector;*
- *project interactive animations and/or images onto a floor;*
- *doesn't have to focus on balancing;*
- *use body movement as 'input';*
- *stimulate social interaction (allowing players/users to cooperate or play together)*

Figure 1: Excerpt from the 2018 Building Engaging Prototypes course assignment: Building a WOz prototype for Practicing Balancing Skills in Physical Education.

Step by Step Wizard of Oz Prototype

Concept	<Name your concept> <Describe your concept>	Example 1: Magnetic Fish Fish follow the user while balancing on the bench, when the user falls off the fish disappear. Example 2: Balloons The amount of projected balloons increase every time the user skips to a new balancing stone
Scenario	What should happen in response to user behaviour? <Describe user behaviour> <Describe system reaction>	Example 1: When the user walks on the bench, the projected fish will follow the user Example 2: For every balancing stone the user jumps on, the amount of projected balloons around the user increase, when the user falls of a stone, he starts from the beginning.
Interface elements	Which interface elements (illustrations, animations, objects) are needed?	Example 1: Fish Example 2: no balloons, 1 balloon, 2 balloons, 3 balloons etc.
Wizard Input	When, Where & How will the Wizard provide input?	Example 1: Wizard will follow user with mouse, fish images follow mouse location in Processing. Example 2: Every time the user jumps to a new balancing stone, the Wizard moves ahead one image (slide) with the arrow keys in Powerpoint. When the user falls of a balancing stone, the Wizard will start the slides/images from the beginning.

Figure 2: Example of step-by-step walkthrough guiding students through the WOz process, part of the WOz assignment.

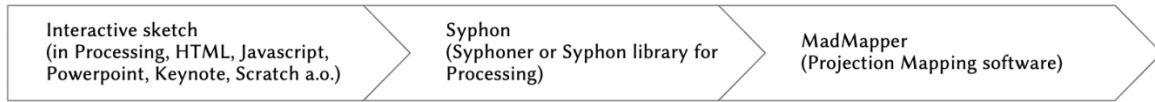


Figure 3: Wizard of Oz Prototyping for interactive Spatial Augmented Reality Workflow

Building Engaging Prototypes (BEP). The emphasis of this course is on letting students experience a variety of prototyping methods and tools hands-on. We should note that the focus is not on the design and development of the products themselves, but on experiencing the prototyping methodology. WOz prototyping is covered during the last two weeks of the course with one introductory lecture to the method and two prototyping workshops, during which the students work on a WOz prototyping assignment.

3.1.1 Lecture. Students start with a lecture about Wizard of Oz prototyping, its history, use in UX prototyping, and best practices for creating and testing WOz prototypes. The assignment is introduced, explaining its context, requirements and constraints (Figure 1) and a step-by-step walkthrough (Figure 2) guiding the students through the prototyping process.

3.1.2 Workshop 1. The first workshop is in the same week as the lecture. In teams of two, students start working on the assignment, come up with a concept for their prototype and discuss their plans with their workshop lecturer. After this workshop, students independently finish the WOz Prototypes as homework before the second workshop.

3.1.3 Workshop 2. The second workshop takes place on the second week and lasts 3-5 hours depending on the size of the group. Students bring their finished prototypes to class. Lecturers have prepared setting up projectors and a computer with projection mapping software (MadMapper and Syphoner) installed, in case students don't have a laptop or are experiencing issues with installing the software.

Depending on the space available, students either test their prototypes sequentially or in parallel. Of each prototyping duo, one student acts as the wizard, while the other facilitates the test and explains the test subject (one of their classmates) what to do. The wizard is not hidden, and all test users are aware of the faked technical backend due to the nature and goal of this workshop. If issues with the prototype occur during the test session, the students are instructed to iterate on their prototype on the spot.

3.2 Assignments

The prototyping assignments all involve some form of interactive spatial augmented reality. The emphasis differs each year and relates to current research goals of the research group Healthy Lifestyle in a Supporting Environment. For example, the assignment in 2017 – 2018 was: *“Create a WOz prototype to support balancing skills. The prototype stimulates social interaction, uses body movement as input and projects (interactive) images or animations onto a floor”*. Appendix A.1 presents a full overview of the assignment topics and constraints from 2017-2022.

3.3 Participants

Between 2016 and 2022, 187 students attended the course, resulting in 93 different WOz SAR prototypes (Table 1).

Table 1: Overview of number of students and WOz SAR projects between 2017 – 2022

Year	Students	WOz SAR Projects
2016 – 2017	30	15
2017 – 2018	34	17
2018 – 2019	40	20
2019 – 2020	32	16
2020 – 2021	16	8
2021 – 2022	35	17
Total	187 Students	93 Projects

3.4 Prototyping Tools

We developed a workflow (Figure 3) to enable students to rapidly create interactive SAR WOz prototypes.

Students create their prototypes in whatever tool they like and feel comfortable with. This can be Processing [16] (Figure 4), an IDE they have previous experience with, or JavaScript and HTML, staying close to the tools with which the interactive products that students often design for are developed with. However, tools are also used that originally have a different purpose but lend themselves to quickly create content that can be controlled by a Wizard, such as PowerPoint or Keynote.

Visual output is then sent through Syphon. Syphon [17] is an open source (Mac OS X) technology allowing real-time sharing of screens between applications. When using Processing, some students use a Syphon library, but nearly all students use Syphoner [18] (Figure 5, Figure 6), an application for screen sharing with Syphon. The reason for needing a screen sharing technology is that for SAR they project on floors, objects, ceiling etc. and therefore need to send their output to projection mapping software, MadMapper [19] (Figure 6, Figure 7). This software enables adjusting the projector output, enabling the digital content to be projected on the projection surface as desired, without distortions.

In cases when students project on a wall, Syphon and MadMapper can be skipped, and a projector can be placed right in front of the wall. However, this may create undesirable occlusion (shadows) caused by the user standing in between the projector and the wall.

4 RESULTS

We reviewed the prototypes that were developed during the span of the past six years to map out their diversity and common elements. We analyzed them based on purpose, input and interaction, play

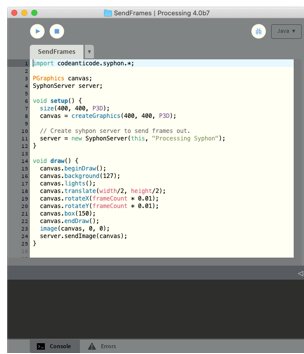


Figure 4: Processing IDE



Figure 5: Syphoner

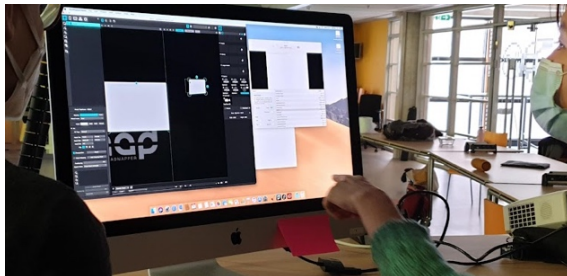


Figure 6: Combining Syphoner with MadMapper

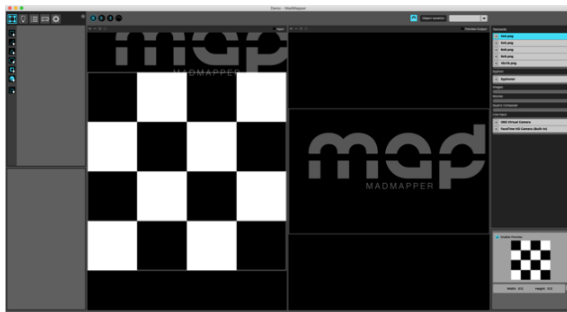


Figure 7: MadMapper Projection Mapping Software



Figure 8: Dinosaur catching falling crabs prototype

characteristics, projection surfaces and dimensions and problems that emerged. In this section, we present the insights gained.

4.1 Purpose

Purpose is one of the features we looked at. This is what the interface contributes to. Our review reveals four main categories in terms of the purpose of the interface: entertainment(games), physical exercise, artistic expression, and task performance. On occasion, these categories overlap, for example when the students have designed a game-like interface for physical exercise (Figure 8).

Entertainment in the form of games has been by far the most occurring category, and within that category there has been a lot of diversity; examples range from well-known game concepts, such as Tic Tac Toe (Figure 9), Twister and catching and/or avoiding falling objects (Figure 8), to more ambitious and adventurous installations such as a duel of wizards casting spells at each other and escaping from a prison by breaking a brick wall.

In the physical exercise category, students have typically designed task-based objectives, such as repeating a certain physical exercise to accumulate points; mimicking yoga postures based on a projected shadow on the wall; and maneuvering through an obstacle course, which also makes use of physical objects such as steppingstones (Figure 10, Figure 11) as part of the installation.

Artistic expression has not been as common as the other categories but has showcased diverse objectives. Students have designed interfaces for visual expression such as drawing shapes on the floor



Figure 9: Tic Tac Toe prototype



Figure 10: Stepping stones balancing game

(Figure 12) but also musical interfaces, e.g., controlling a sampler by hopping on and off projected “tiles”, and mixing pre-recorded music with a wall-projected visual interface (Figure 13).

Prototypes that focus on task performance include interfaces for tasks such as. gesture based interaction for browsing video streaming services and the aforementioned music making (Figure 13).

4.2 Input & Interaction

In their prototypes, students have applied a very broad variety of possible ‘inputs’ to the system, which have been tracked by the wizard in the experiment. Due to the nature of the assignment context, most installations have been designed around movement and motions as the primary input; for example, stepping on or touching the floor (Figure 10, Figure 11); tapping or touching a projected object; various gestures (dragging, holding, throwing, pointing);



Figure 11: Stepping stones balancing game



Figure 12: Floor painting interface

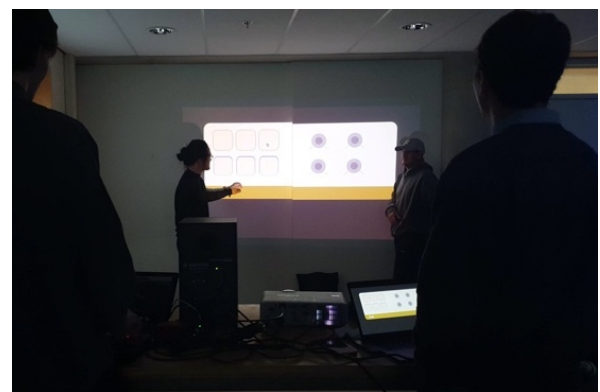


Figure 13: Wall music making interface



Figure 14: Player using a toy spade as input object

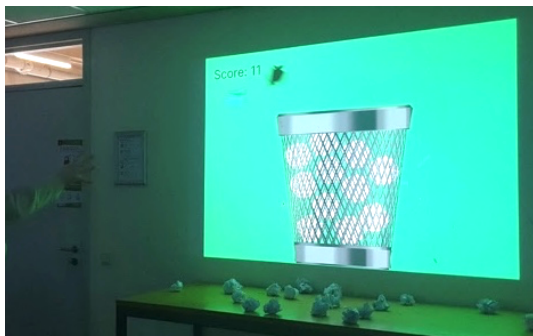


Figure 15: Paper trash as input objects

and full body movement in or around the area of projection. In addition to movement, some students have applied proxy objects such as controllers or real objects as the input, such as a spade (Figure 14) and paper trash (Figure 15).

4.3 Play Characteristics

Playful interfaces can be defined by various characteristics. In our analysis we could divide the prototyped interfaces according to the following characteristics: number of players, ending and objective.

Most prototypes were single player interfaces (Figure 8, Figure 10, Figure 11, Figure 13, Figure 14, Figure 15), some were specifically for two players (Figure 16) and others allowed for more simultaneous players (Figure 17, Figure 18). For the wizard, tracking just one user was often easiest. In cases where the wizard was merely responsible for indicating whether the players' moves were correct or faulty this was just as doable.

The prototypes had various objectives. Most were competitive, having to reach a goal (Figure 14) or beating a previous player as the objective. Some were cooperative, where players had to help each other in reaching a goal, such as matching a shape together (Figure 16), touching all the projected dots (Figure 17) or matching the numbers of a clock floor projection (Figure 18).

4.4 Projection Surfaces

The interface prototypes used different surfaces to project onto: floors (Figure 8, Figure 9, Figure 13, Figure 18) (also because in some



Figure 16: Cooperative play with two players



Figure 17: Cooperative play with three players

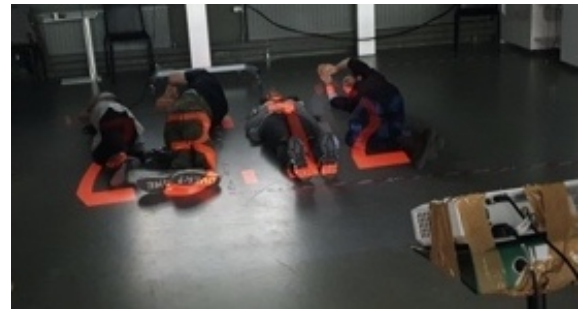


Figure 18: Cooperative play with four players

editions this was a requirement in the assignment), walls (Figure 8, Figure 9, Figure 13, Figure 14, Figure 15, Figure 16, Figure 17), tables (Figure 19), objects (Figure 20) and the ceiling (Figure 21). Surfaces were used in a variety of different sizes and for different purposes. It can be seen that the constraint of the projection surface was used creatively by students by choosing different sizes, different angles and contexts for their interactions.

4.5 Insights

As is often the case in early prototypes, the students have encountered various problems in the testing sessions. These problems can be divided into three broad categories: problems with the setup, problems with the prototype, and human factors.



Figure 19: Table projection



Figure 20: Wall projection combined with real object



Figure 21: Ceiling projection



Figure 22: WOz prototype of Shape and Shadow Matching Game

4.5.1 Projector setup. In terms of the setup, a prominent issue is occlusion. Especially with floor projections, it is common that the people interacting with the installation are partially occluding the projection. While this can be partially remedied by using a short-throw projector, placing the projector in a high-enough angle and letting the user move towards the projector, in certain installations the occlusion can have a major impact on the experience, especially if the people are supposed to interact with the installation by moving in the projected area. The same phenomenon can also occur with wall projections, although then it is easier to mitigate with projector placement. This is however a problem that is not limited to the Wizard of Oz prototype and will also occur with ‘real’ SAR interfaces and is therefore helpful in gaining insight in designing for these types of interfaces.

4.5.2 From flaw to feature. Limitations were sometimes used intentionally, sometimes unintentionally, in a creative way to enhance the interaction. The shadow caused by people or objects in front of the projector, for instance, was often utilized by students to perceive where someone touched, tapped, or moved. Such as in the shadow matching game (Figure 22) where players had to match a projected shape with their shadow.

4.5.3 Software issues. Another technical issue is that the Syphon software used is not fully compatible with all kinds of source software and only works on Mac OS. Because we are not limiting the students in terms of which software platform they use to realize the prototype output, there have been some unexpected issues with latency, for example, when the prototype was made with a presentation tool (PowerPoint, Keynote). In these cases, the projection does not respond quickly enough to the output software.

4.5.4 Contrast. A typical problem with the prototypes is contrast. While we aim at making the space as dark as possible for the test session, there is always some natural light and many of the surfaces are white or light grey. This sets limitations for the colors that the students can use in the visual elements of the prototype to maintain intelligibility of the projection. In a typical test session,

some students need to tweak on their prototype to maximize the contrast of the projection. Although this too is a limitation that will also occur with ‘real’ interfaces and provides the students with valuable insights to take into consideration in their designs.

4.5.5 Wizarding. Finally, human factors come to play when the wizard is controlling the output of the experiment. Especially with more complex prototypes in terms of their input commands but also with fast-paced interactions, the wizard cannot always keep up with the physical actions that the test participants undertake. This can result in delayed response, incorrect response, or inaccurate response for example if the tracking has designed to be very fine-grained.

In the prototype test sessions, the wizard was not hidden (as is normally the case in WOz prototyping). Students acted as each others test users and were well aware of the fake backend. Still interaction with the prototypes appeared highly immersive and knowing users participated in the interfaces as if they were fully working.

5 DISCUSSION

The WOz prototyping assignment has proved to be a powerful educational tool. As can be seen from the analysis of the prototypes, the educational approach stimulates the students to explore and experience the design space and possibilities of SAR applications. The diversity in interaction types and objectives of the interfaces suggests that the method facilitates the creativity of the students. While certain patterns in the content of the prototypes seem to repeat on a yearly basis (variations of familiar concepts such as Twister and lava floors), every run of the course brings forth also more exploratory, unexpected, and ambitious prototypes.

Some qualities of WOz prototyping turned out specifically helpful for the assignments: The quick, iterative approach that is made possible by focusing on simpler technical solutions led to more open and free thinking among students. This also led to more focus towards the design of the actual interaction, rather than spending time on the technical setup. It can be observed in the examples that students spend most of the time designing interactions, rather than implementing a technical setup.

Furthermore, the constraints of the assignment and the WOz prototyping methods were used creatively by students (see 4.5.2.). Beyond that, the method elicited creative thinking in students, which can be observed by the multitude of different applications and interactions developed by the students.

It can be argued that the aim of introducing WOz as a low-barrier prototyping method for technically demanding interactive products is met in the prototyping course. The total time for contact teaching in this assignment is approximately 8 hours (one lecture + two workshops), which has proven sufficient for the students to both understand how the prototyping method works and to create an experiential prototype that goes beyond the GUI interfaces they have previously designed in their studies.

One of the constraints of the WOz prototyping method is at the same time its strength: the exclusion of implementation. While this leads to quick and imaginative solutions, this can also lead to unrealistic prototypes, that are not possible to implement. This fact

might lead to frustrations in students, who are eager to take their prototype to the next step.

Furthermore, the proposed workflow and use of tools also leads to results that favor specific kind of interactions, because the tools suggest specific affordances. Especially, when using those tools unintendedly (e.g., Powerpoint animations for moving objects on a wall) some interactions are easier to realize. This can be seen in some examples: many students use interactions that focus on one moving object, because it is easy to fake this with one mouse input for the wizard. If the tools allowed for multiple mouse inputs for instance it is likely that prototypes would have gone in different directions.

In addition to the outcomes of the WOz assignment itself, many students make use of the WOz approach in their projects in the later study years. The low-cost WOz approach lets the students try out their designs on location with test participants from their target group. As an example, in a second-year project the students design an interactive system for a shared space, such as a museum, library, or gym hall, in which they often use WOz prototyping to test and iterate on their design.

In the design process, WOz prototyping could be useful to gain more imagination of what an interactive system should be able to do. Beyond developing new ideas for interactions and SAR, WOz could help imagine and discuss a potential technical implementation. However, based on the data of this study it is not possible to conclusively say what imaginations are triggered in the students through the workshop. This case study is mostly relying on documentation, observation and experience of the facilitators, but could furthermore be extended to include student’s perspective from the workshop.

6 CONCLUSION

Based on the analysis of the 93 prototypes created between 2016 and 2022, and our experiences during the workshops, we conclude that the WOz workshops stimulate creativity, enable students to rapidly create interactive SAR interfaces without being limited by their technical skills and time limitations and provide them with a tool that they apply beyond the scope of the prototyping course. The workshops have resulted in a variety of prototypes and these provide insight into the advantages and limitations of using WOz prototyping for designing interactive spatial augmented reality.

ACKNOWLEDGMENTS

We thank all CMD UXD students at the Hague University for Applied Sciences who took part in the Building Engaging Prototyping course between 2016 and 2022. We also thank our (former) colleagues William Beekhuis, Ruud Brok, Kees Sommer and Kristel Kerstens for teaching this course with us.

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A APPENDICES

A.1 Assignments and requirements from 2017-2022

WOz prototyping assignments and requirements between 2017-2022

Year	Assignment	Requirements / Restrictions
2016 - 2017	Build a Wizard of Oz prototype for BalanSAR. (. . . that supports practicing balancing skills in Physical Education)	•work with a video projector, floor projection and either balancing stones or bench. •doesn't have to focus on balancing. •use body movement as 'input'
2017 - 2018	Create a WOz prototype to support balancing skills. The prototype stimulates social interaction, uses body movement as input and projects (interactive) images or animations onto a floor.	•work with a video projector. •project interactive animations and/or images onto a floor •doesn't have to focus on balancing. •use body movement as 'input' •stimulate social interaction (allowing players/users to cooperate or play together)
2018 - 2019	Come up with a new concept for an exertion interface (using bodily movements as input) that stimulates/ supports/ evokes/ nudges social interaction and uses projection mapping (aka. Spatial Augmented Reality - SAR)	•work with a video projector. •project interactive animations and/or images onto a floor, walls, objects or a combination. •doesn't have to focus on balancing. •use body movement as 'input' •stimulate social interaction (allowing players/users to cooperate or play together)
2019 - 2020	Come up with a new concept for an exertion interface (using bodily movements as input) that stimulates/ supports/ evokes/ nudges social interaction and uses projection mapping.	•work with a video projector. •project interactive animations and/or images onto a floor, walls, objects or a combination. •use body movement as 'input' •stimulate social interaction (allowing players/users to cooperate or play together)
2020 - 2021	Create and test a prototype for a new concept for a playful social distancing interface that uses projection mapping and (faked) camera and/or other sensory input.	•work with a video projector. •project interactive animations and/or images onto a floor, walls, objects, people or a combination. •use body movement/presence as (faked) 'input'.
2021 - 2022	Create and test a prototype for a new concept for a playful interface that uses projection mapping and (faked) camera and/or other sensory input. We recommend focusing on evoking a playful interactive experience; that doesn't mean it has to be a game. Try something open-ended and unstructured.	•work with a video projector. •project interactive animations and/or images onto a floor, walls, objects, people or combination. •use body movement/presence as (faked) 'input'.